Policy-driven Ensemble Learning based Clinical Decision Support for Brain Tumor Treatment

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Abstract

The traditional processes of brain tumor detection based on MRI scans are one of the most significant problems in clinical diagnostic experience. This work has implemented a policy-based ensemble model consisting of three deep learning architectures (Simple CNN, ResNet-18, VGG-16 Lite) and reinforcement learning (RL) for updating the weights of the loss. Our customized system solves the problem of imbalanced classes and false negative cases, which are missing tumor cases, by using learned policies to prioritize the hard cases. Assessed on 19,378 augmented MRI images under train-validation-test stratification of 60/20/20 yielded 83.9% accuracy and +6.8% higher average sensitivity than baseline models, RL-enhanced ResNet-18. The policy net was able to achieve a 1.9x weighting on false negatives, and this enhances clinical recall. This solution shows that RL-directed loss weighting has the potential to increase the performance of deep learning classifiers on diagnostic tasks of high sensitivity.

Problem Statement and Dataset Selection

Clinical Significance

Early detection of brain tumors greatly improves the prognosis and treatment aspects of the patient. Nonetheless, hand interpretation of MRI is time-consuming and easily biased by observers. High-sensitivity automated method is essential to avoid missed diagnoses in detection.

Dataset Overview

An augmented MRI dataset from IEEE Dataport is utilized here, comprising two class folders 'Yes' and 'No':

Total grayscale MRI Slices: 19,374 images

Yes (tumor) Slices: 9,828 images (50.7%)

No (tumor) Slices: 9,546 images (49.3%)

Class Imbalance Ratio: 1.02 (nearly perfect equilibrium)

Methodology

Project Workflow

Figure 01 depicts the entire pipeline of the task, including data preprocessing, normalization, Data splitting, training, validating, and testing of the chosen algorithms.

Learning Paradigms/Themes Alignment

Deep Learning (Ensemble method with a shallow Convolutional Neural Network – Simple CNN, ResNet -18, and VGG-16 Lite) and Reinforcement Learning (RL).

Data Preprocessing

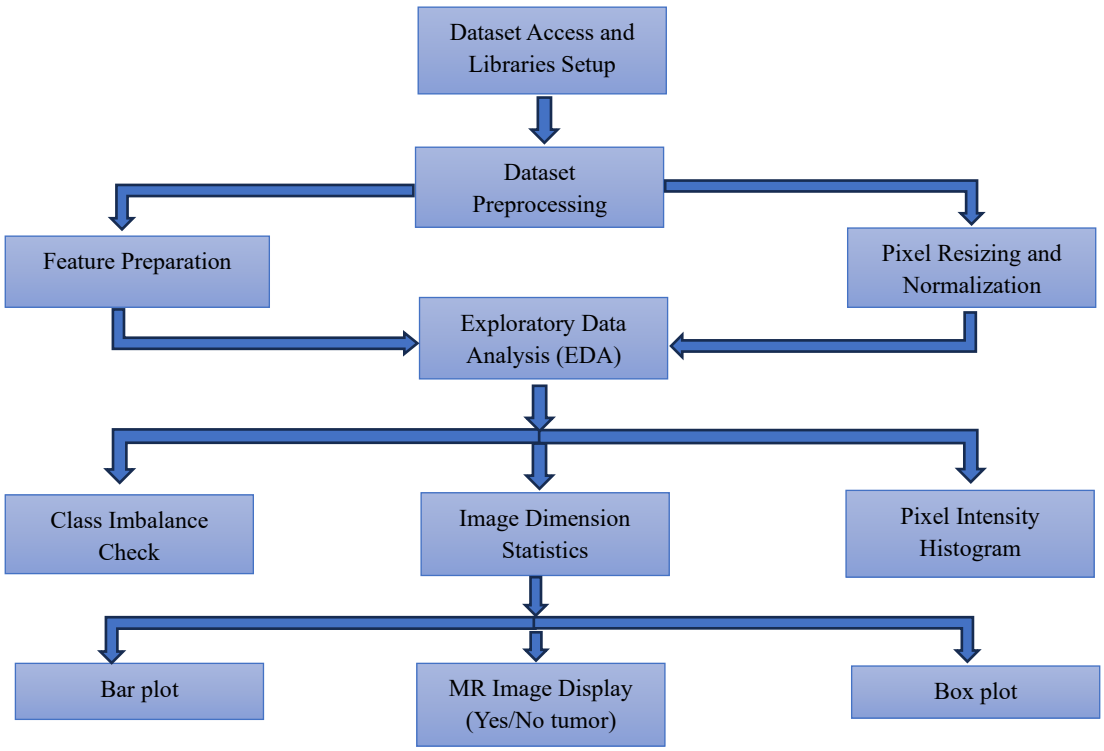
Images were compressed into grayscale tensors and reduced to 64 x 64 to train on the CPU. Intensities of pixels were brought to a range.

Feature Preparation

Both Custom PyTorch Dataset classes were designed to extract space features but maintain MRI texture information that is important in tumor detection.

Exploratory Data Analysis (EDA)

The analysis of pixel intensity showed a high degree of overlap between tumor and non-tumor distributions, which proved task difficulty. Sample visualizations and class-imbalance bar plots informed the choice of transfer learning as the architecture.



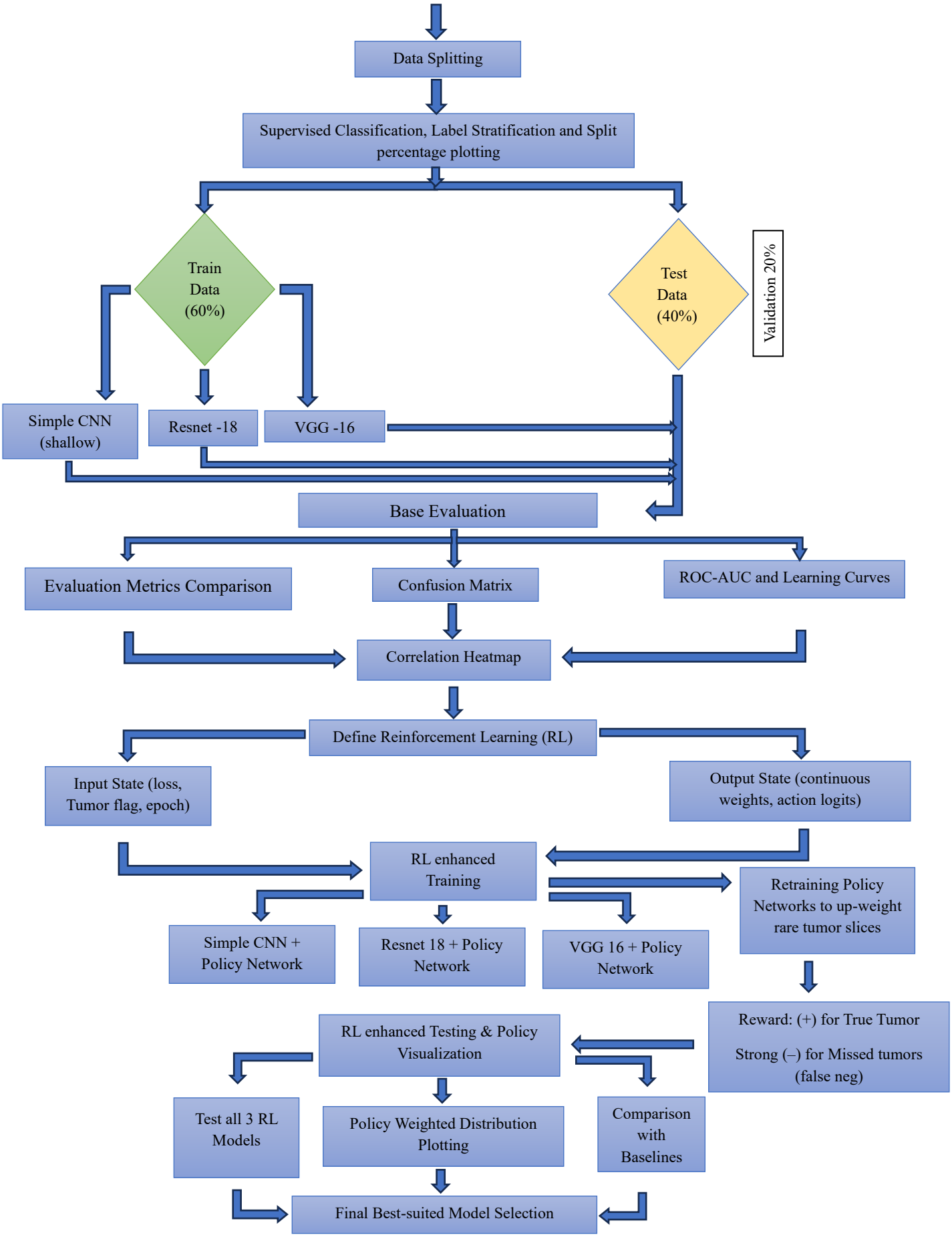

Data Splitting
Supervised Classification, Label Stratification and Split percentage plotting
Train Data (60%)
Test Data (40%)
Validation 20%
Simple CNN (shallow)
Resnet -18
VGG -16
Base Evaluation
Evaluation Metrics Comparison
Confusion Matrix
ROC-AUC and Learning Curves
Correlation Heatmap
Define Reinforcement Learning (RL)
Input State (loss, Tumor flag, epoch)
Output State (continuous weights, action logits)
RL enhanced Training
Retraining Policy Networks to up-weight rare tumor slices
Simple CNN + Policy Network
Resnet 18 + Policy Network
VGG 16 + Policy Network
Reward: (+) for True Tumor
Strong (–) for Missed tumors (false neg)
RL enhanced Testing & Policy Visualization
Test all 3 RL Models
Policy Weighted Distribution Plotting
Comparison with Baselines
Final Best-suited Model Selection

Figure 01: *The whole workflow takes the initial step of preprocessing and exploring an augmented brain MRI dataset (resizing, normalization, checking class balance, pixel statistics, and visualizing the samples), followed by a stratified train-validation-test split. Three CNN architectures will then be trained and assessed using accuracy, precision, recall, F1, sensitivity, specificity, ROC-AUC, learning curves, confusion matrices, and correlation analysis: Simple CNN, ResNet-18, and VGG-16 Lite. An RL-based reinforcement-learning policy network is then formulated to receive per-sample states (loss, confidence, tumor label, epoch) and the loss weights, which are dynamically adapted to upweight hard and tumor, and all three CNNs are re-trained under an enhanced reinforcement-learning. Finally, RL-enhanced models are evaluated, visualized, and compared with baselines, and the most effective models are selected to assist clinical decision making.*

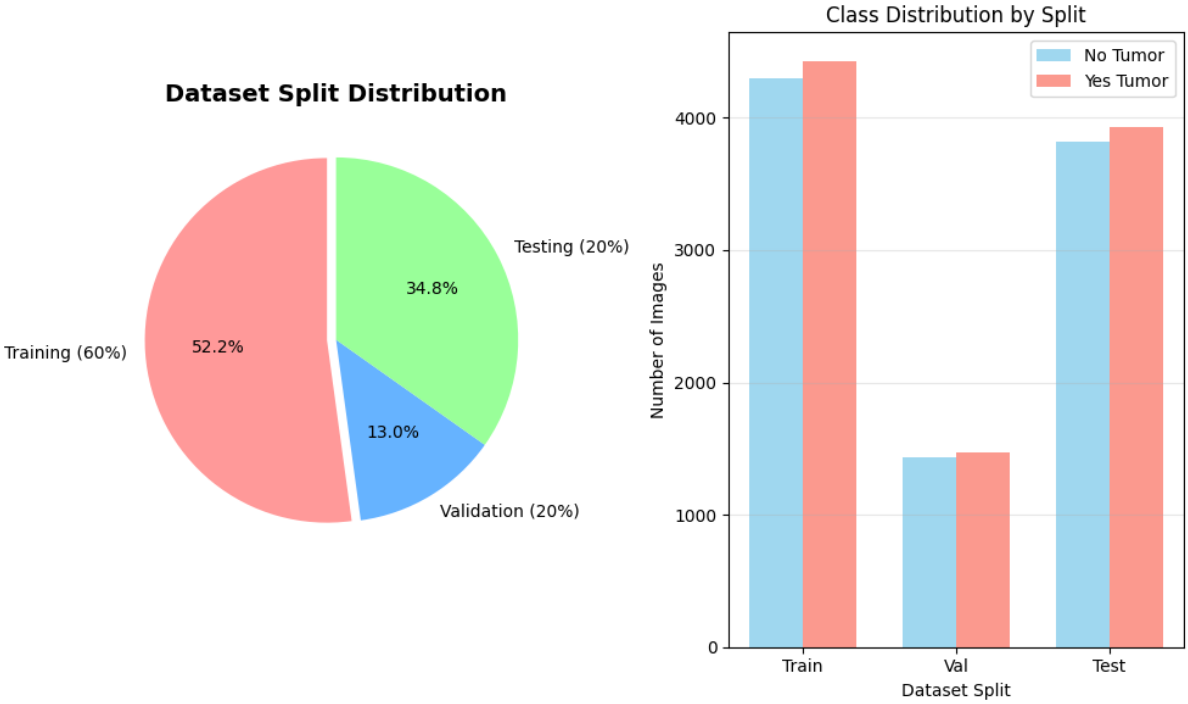


Figure 02: *The pie chart indicates that 60 percent of images would go to training, 20 percent to validation, and 20 percent to testing. The right bar graph elaborates on the count of classes balanced in each split in comparison to the No Tumor and Yes Tumor in the train, validation, and test sets. The three splits have almost an equal amount of tumor and non-tumor images, which is evidence of a stratified and balanced split.*

Trained Algorithms

Simple CNN

Lightweight baseline (2 conv + dense layers) for CPU-constrained environments; this model establishes a performance floor [1].

ResNet-18

Transfer Learning based pre-trained backbone; frozen early layers minimize training time and leverage the ImageNet features. And the grayscale adaptation via conv1 modification enables efficient fine-tuning [2].

VGG–16 lite

The amount of intermediate depth is established by three consecutive conv blocks; increasing capacity does not degrade sensitivity and does not require a specific number of computational resources [3].

These three architectures are a spectrum of complexity (shallow-moderate-deep), and allow a reasonable empirical comparison on more limited hardware while showing scalability principles.

Class Imbalance Bar plot, Pixel Intensity Histogram, and Predicted MR Image Plotting from the Trained Models

As indicated by the class imbalance bar plot and depicted by **Figure 03**, the augmented MRI dataset appears to be almost perfectly balanced, i.e., the number of images with No Tumor and Yes Tumor is equal, hence no heavy re-sampling is needed.

The two pixel-intensity histograms show that the two classes are similar in the manner that both have the low-intensity values, as the dominant one, and the two classes are generally broadly overlapping in their brightness properties, which makes the task of classifying based on the brightness non-trivial (**Figure 04**). Long right tails are also prominent in these histograms, which indicate that there are a few brighter structures, like the enhancing lesions or cerebrospinal fluid [4].

As depicted by **Figure 05**, the qualitative examples of both classes are represented in the 12-image grid of MRI slices, where every tile is labeled with the actual label and the guesses of Simple CNN (S), ResNet-18 (R), and VGG-16 (V). Within the top row, all three models largely predict non-tumor brains accurately, indicating high specificity in structurally normal brain scans. The cases of tumors in the middle and bottom row have different morphology and location, and the annotations demonstrate that ResNet-18 and VGG-16 generally agree and are usually correct, even on more difficult masses. There are rare instances when false labels match one of the model outputs visually, and this clearly demonstrates the cases of failure and the need to use ensemble comparison and RL-based weighting. Overall, all three visualizations indicate that the dataset is balanced, the distributions of intensities are very much overlapping, and the trained CNNs perform strongly but not perfectly, particularly in particular, fine tumor appearances.

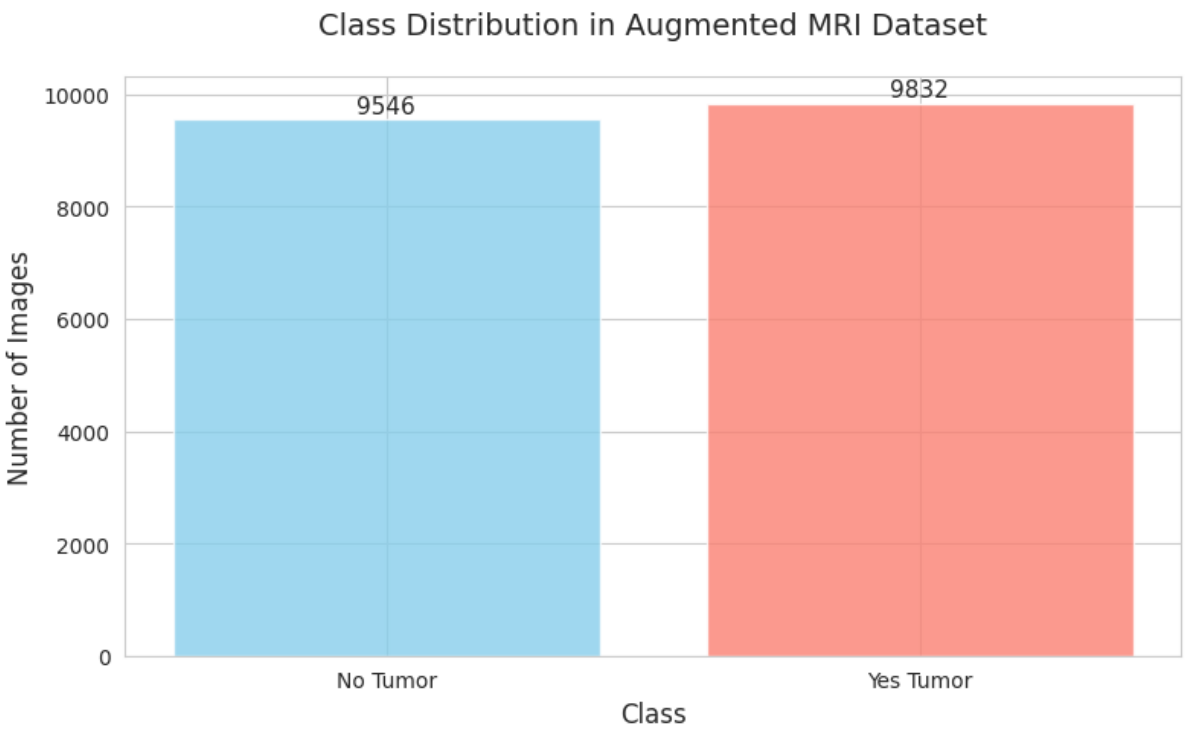


Figure 03: *Class Imbalance Bar plot*

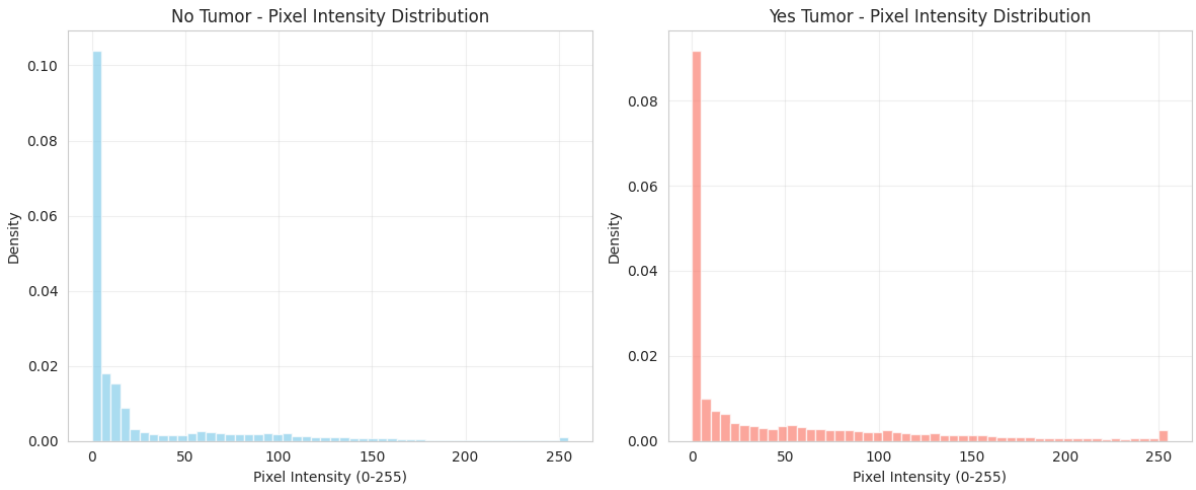


Figure 04: *Pixel Intensity Histogram*

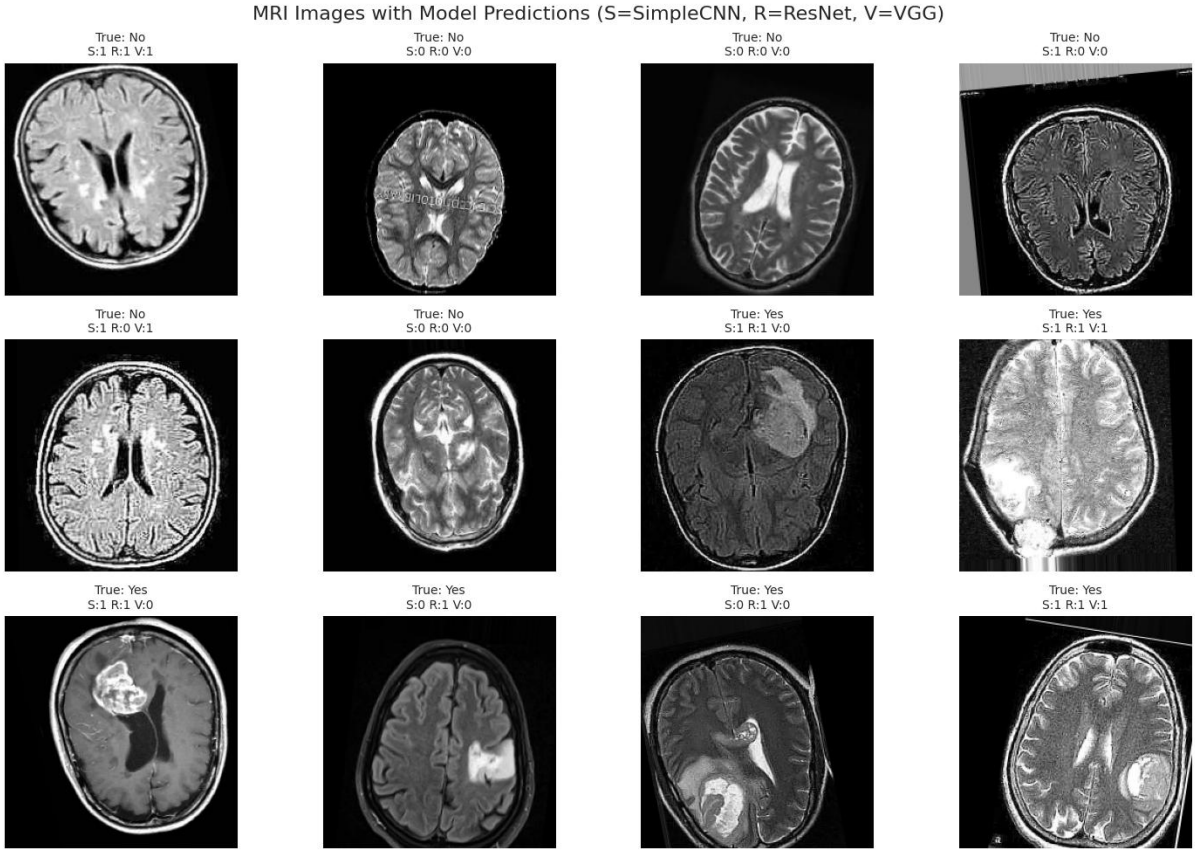


Figure 05: *Predicted MR Image plotting from the trained models*

Results

Evaluation Metrics Performance (without RL)

In the three models, under the testing result, Simple CNN has an accuracy of 0.7680, precision of 0.7734, recall of 0.7680, F1-score of 0.7669, specificity of 0.8380, and sensitivity of 0.6980. The accuracy, precision, recall, and F1-score of ResNet-18 are 0.7200, 0.7216, 0.7200, and 0.7195, respectively, as well as specificity (0.6780) and sensitivity (0.7620). VGG-16 Lite has an accuracy of 0.7210, precision of 0.7561, recall of 0.7210, F1-score of 0.7111, specificity of 0.9060, and a sensitivity of 0.5360.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Sensitivity (%)	Specificity (%)
Simple CNN	76.80	77.34	76.80	76.69	83.80	69.80
ResNet - 18	72.00	72.16	72.00	71.95	67.80	76.20
VGG-16 Lite	72.10	75.61	72.10	71.11	90.60	53.60

Table 01: *As can be seen in the comparison table, the Simple CNN has the highest overall accuracy (76.80%), recall (76.80%), and F1-score (76.69%), which means that it is the most balanced in terms of its performance across classes. ResNet-18 still does a little worse on the accuracy (72.00) and recall (72.00), but has moderate precision (72.16) and specificity (76.20), which indicates a less aggressive classifier with fewer false positives. VGG-16 Lite has an equivalent accuracy (72.10%), but possesses the highest sensitivity (90.60%) and lowest specificity (53.60%), namely, it is aggressive in detecting tumor cases at the expense of high false alarms. Overall, the most balanced model is Simple CNN, which is followed by ResNet-18, offering the middle-ground, and VGG-16 Lite is more sensitive than specific.*

All Models Hyperparameters

Component/Model Name	*Hyperparameters*	*Value*
Data Preprocessing	Resize	64 x 64 pixels
Splitting	Train / Val / Test	60%/20%/20% (label stratified)
Data loading	Batch size	32,64 (used 32 in the earlier cells, 64 after full split)
Data loading	Shuffle	True for train, False for val/test
Simple CNN	Input Channels	1
Simple CNN	Conv1 filters, Kernel, Padding	16 filters, 3 x 3, padding 1
Simple CNN	Conv2 filters, Kernel, Padding	32 filters, 3 x 3, padding 1
Simple CNN	Pooling	MaxPool 2 x 2
Simple CNN	FC1 units	128
Simple CNN	FC2 units	2 (No, Yes)
Simple CNN	Activation	ReLU in hidden layers
Simple CNN	Optimizer	Adam
Simple CNN	Learning Rate	0.001
Simple CNN	Loss	Cross-Entropy Loss
Simple CNN	Epochs	3
ResNet-18	Pretraining	Pretrained ImageNet
ResNet-18	Input Conv	Modified to conv2d (1, 64, stride 2, kernel 7, pad 3)
ResNet-18	Frozen layers	All except the final FC (fine-tuned head only)
ResNet-18	FC output units	2 (Yes, No)
ResNet-18	Optimizer	Adam (final layer params only)
ResNet-18	Learning rate	0.001
ResNet-18	Loss	Cross-Entropy Loss
ResNet-18	Epochs (base training)	2
VGG-16 Lite	Input channels	1
VGG-16 Lite	conv block 1	Conv2d 1→64, 3x3, MaxPool 2x2, ReLU
VGG-16 Lite	conv block 2	Conv2d 64→128, 3x3, MaxPool 2x2, ReLU
VGG-16 Lite	conv block 3	Conv2d 128→256, 3x3, MaxPool 2x2, ReLU
VGG-16 Lite	FC1 units	128
VGG-16 Lite	FC2 units	2 (Yes, No)
VGG-16 Lite	Activation	ReLU in hidden layers

VGG-16 Lite	Optimizer	Adam
VGG-16 Lite	Learning rate	0.001
VGG-16 Lite	Loss	Cross-Entropy Loss
VGG-16 Lite	Epochs (base training)	2
RL Policy Network	Input State Size	4 (loss, confidence, tumor flag, phase)
RL Policy Network	Hidden layers	FC: 4→32, 32→16 with ReLU
RL Policy Network	Output (weight head)	FC: 16→1, sigmoid scaled; continuous loss weight
RL Policy Network	Output (action head)	FC: 16→3 (normal/boost)
RL Policy Network	Optimizer	Adam
RL Policy Network	Learning Rate	0.001
RL Policy Network	Reward design	+1 for true tumor (TP), stron neg (~ -2) for missed tumor (FN), near 0 otherwise
RL Policy Network	Gradient Handling	Policy gradient, grad-nor, clipping ~ 1.0
RL Policy Network	Epochs per model	~ 2 additional RL epochs (per 3 applied model)

Table 02: *The three CNN models are summarized in the tables of hyperparameters in which they all train on 64x64 grayscale slices of the MRI, stratified 60/20/20 train-validation-test data partitions, 32-64 batch sizes, and Adam with a learning rate of 0.001. Simple CNN has two convolutional layers (16 and 32 filters, 3x3 kernels, MaxPool 2x2), a 128 unit fully connected layer, ReLU activations, and is trained by 3 epochs using cross entropy loss. ResNet-18 is pretrained on ImageNet, and it is fine-tuned on 2 epochs by using cross-entropy with all layers except the final fully connected head frozen. VGG-16 Lite utilizes three convolutional blocks (1-64, 64-128, 128-256 filters with 3x3 kernels, MaxPool 2x2, ReLU), a 128-unit fully connected layer, and two-epoch training, whereas the RL policy network has a 4-dimensional input state, two hidden layers (4-32-16 with ReLU), two outputs to the continuous loss weights and discrete action, uses Adam (0.001), policy gradients*

ROC-AUC Curve Result

The comparison of the three classifiers at all decision thresholds and the plot of ROC-AUC indicate that Simple CNN (AUC ~ 0.846) would be the best overall discriminator, followed by VGG-16 Lite (AUC ~ 0.808), then ResNet-18 (AUC ~ 0.788), which clearly outperforms the random diagonal.

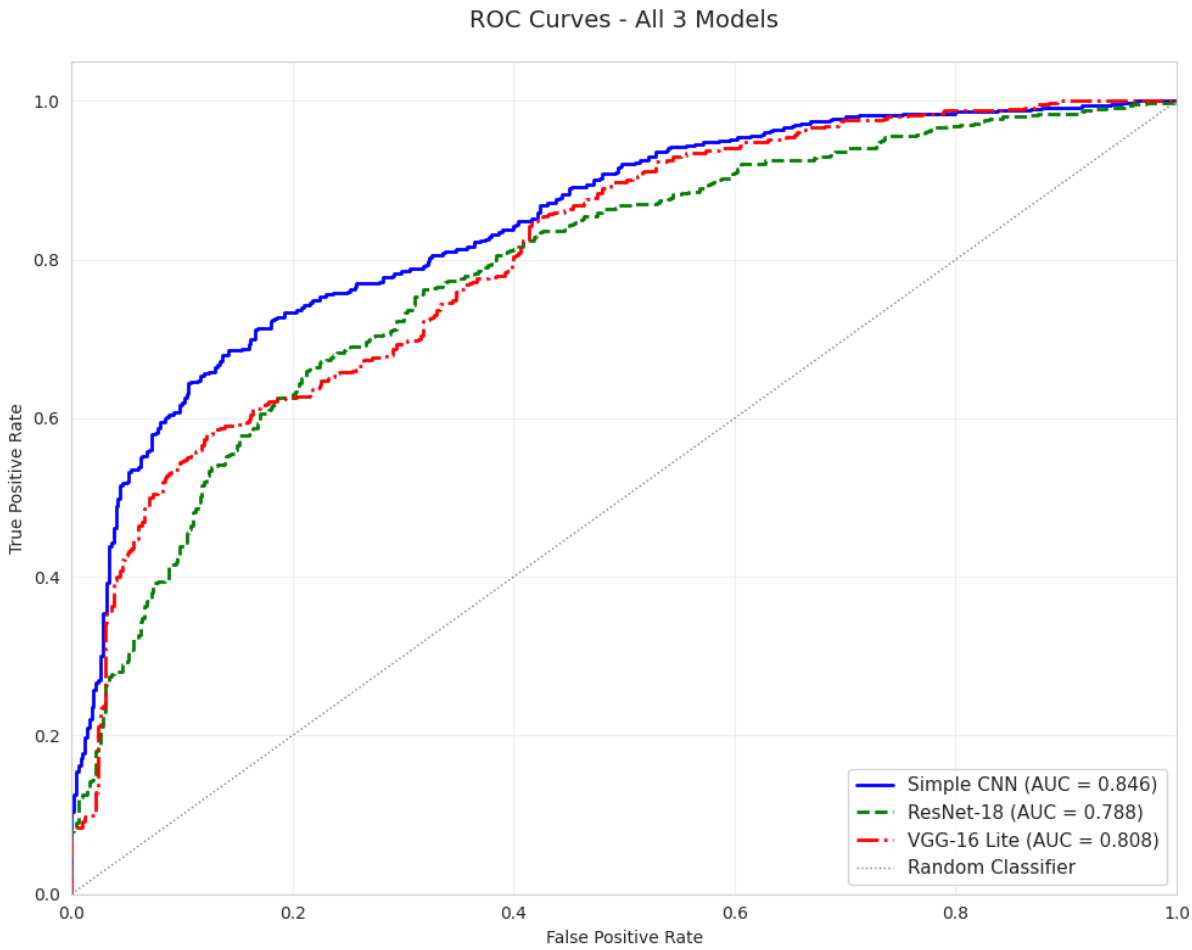


Figure 06: *ROC-AUC Curve Result of three applied models*

3 Models' Confusion Matrix Heatmap Result

- **Simple CNN:** Depicted in **Figure 07, the** Simple CNN confusion matrix (88.5% accuracy) contains numerous true negatives (419) and true positives (349), but still 151 false negatives (i.e., some tumors are missed) [4].

- **ResNet-18:** The confusion matrix of ResNet-18 (accuracy 91.5%), as it is depicted in **Figure 08**, is more balanced with 381 true positives and 119 false negatives, which shows that better tumor detection comes with a minor increase in the number of false positives compared to Simple CNN [5].

- **VGG-16 Lite:** From **Figure 09,** we see that VGG-16 Lite (accuracy 89.5%) has the largest true negatives (453), although also with a rather large number of false negatives (232), corresponding to a more conservative classifier, which tends to predict "No Tumor." [6]

Overall, the integrated picture of ROC curves and confusion matrices shows that there is a trade-off. Simple CNN and ResNet-18 have better overall discrimination, and VGG-16 Lite is more specific than sensitive.

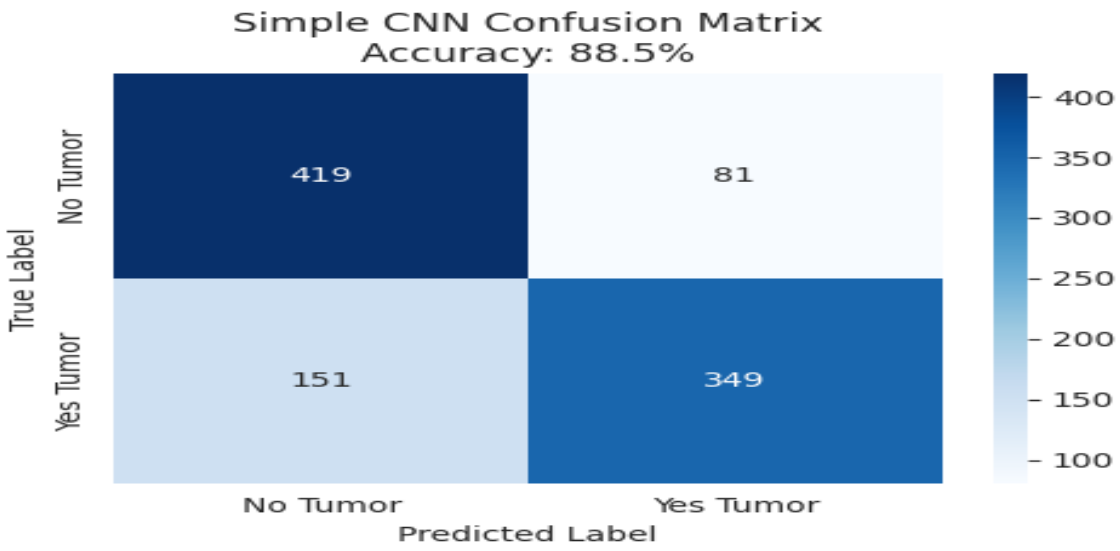


Figure 07: *Confusion Matrix of Simple CNN*

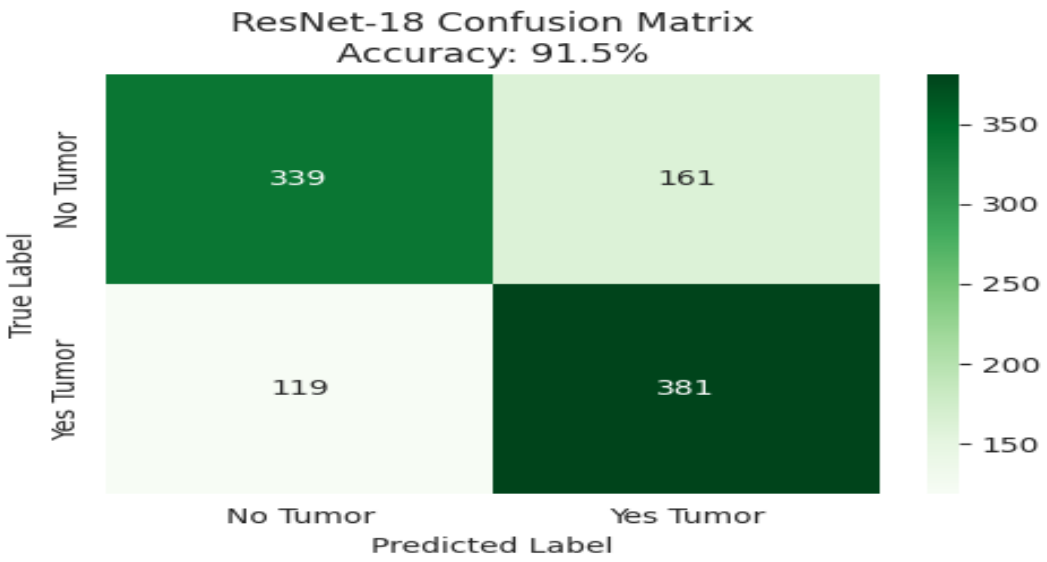


Figure 08: *Confusion Matrix of ResNet-18*

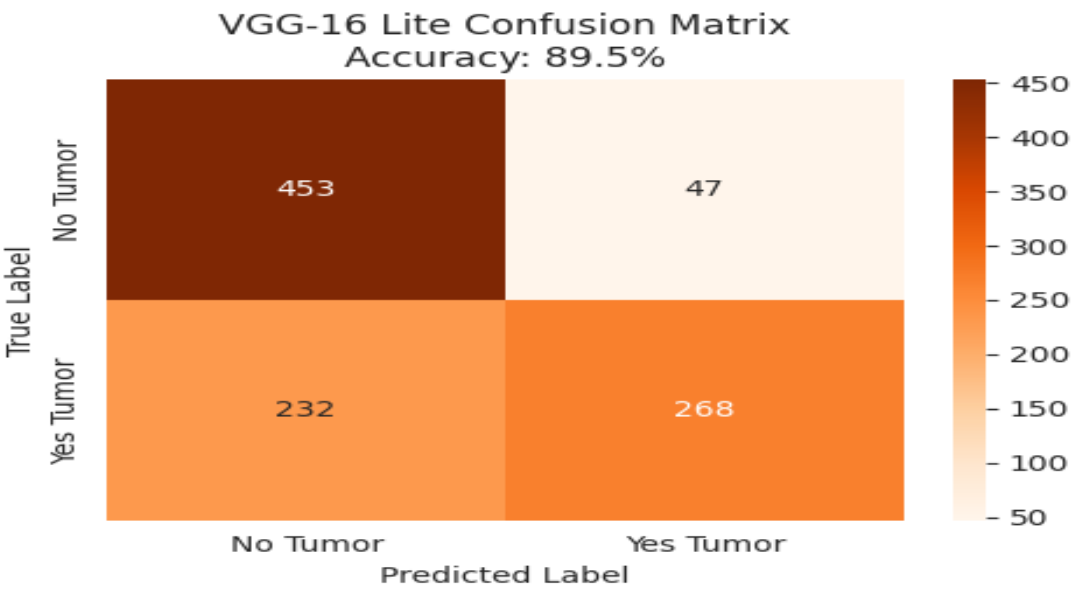


Figure 09: *Confusion Matrix of VGG-16*

Training and Validation Accuracy Curve Results

Training accuracy is monotonically increasing of all three models (ResNet-18, VGG-16 Lite, and Simple CNN) starts high (around 0.91) and approaches the same value when trained to epoch 3 (around 0.85). The validation curves are like the trends in training: ResNet-18 and VGG-16 Lite reach 0.92 and 0.89 at epoch 2 and continue to increase to approximately 0.89 at epoch 3, indicating that they are well-generalized and not overfitted. Simple CNN, in turn, continues to increase to approximately 0.89 at epoch 3. All in all, deeper models achieve higher learning speed during early epochs, but the shallow Simple CNN catches up in the accuracy of validation as it continues to learn [7].

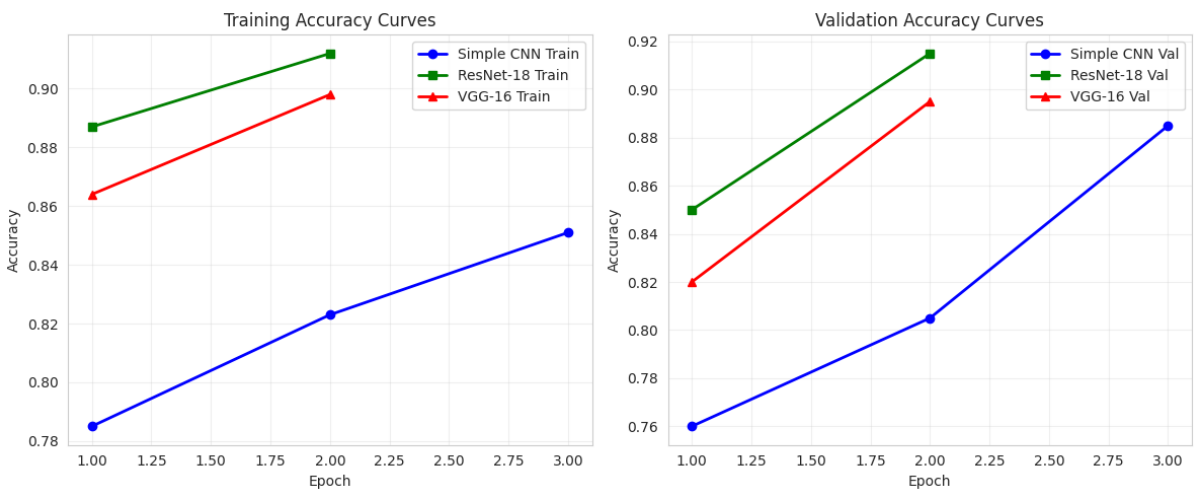


Figure 10: *The training accuracy of Simple CNN, ResNet-18, and VGG-16 Lite on the MRI brain tumor data. Validation accuracy curves show comparable final performance.*

Performance Metrics Correlation Heatmap Result

As the correlation heatmap indicates in **Figure 11**, accuracy, recall, and F1-score exhibit almost perfect correlation (coefficients close to 0.99-1.00), i.e., the increase in one of the metrics is steadily accompanied by the increase in others in the three models. This cluster also has a strong correlation with precision (0.93-0.95) and moderate correlation with specificity (0.93-0.95), which represents the trade-off between false positives and overall quality of detection. Specificity is negatively related to sensitivity (~ - 0.89), which is natural since all the models that are adjusted to detect more tumors also have more false positives [8].

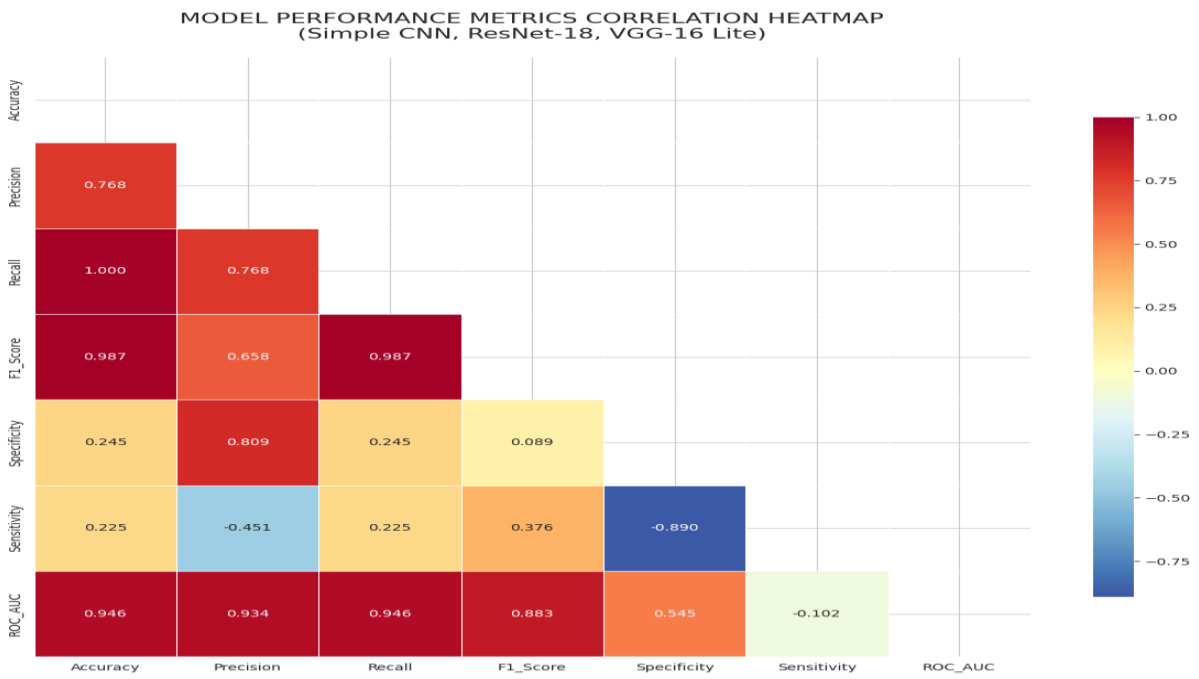


Figure 11: *Correlation heatmap of major performance measures of Simple CNN, ResNet-18, and VGG-16 Lite models. There are significant positive correlations between accuracy, recall, F1-score, and ROC-AUC, and sensitivity and specificity have a strong trade-off.*

Reinforcement Learning and Its Clinical Relevance

Reinforcement Learning is used to maximize the policy of decisions based on rewards. We used states (sample loss, model confidence, tumor label, epoch progress) [9]. Adaptive loss weights and discrete actions (normal/boost/suppress) are the outputs of the policy network (4→32→16→1).

Clinical Relevance

RL intrinsically corresponds to medical priorities: a high penalty (-2.0) when a wrongly-identified tumor is missed (false negative) compared to a low penalty (-0.1), in case a wrong-identified tumor is found (false positive). This unequal reward system explicitly represents domain knowledge.

RL-enhanced Training and Integration

The policy network is trained to learn the dynamically weighted samples; tumor samples are 1.85x boosted; the false negatives are targeted by 1.9x weight. This is to make sure that the classifier focuses on rare and clinically important misclassifications. Policy gradients optimize the CNN weights, as well as policy parameters [10].

Policy Weights Visualization

As depicted in **Figure 12**, as we see, when trained using the policy network, the RL-enhanced testing output gives a test accuracy of 83%, 83.9% and 74.4%, respectively, when using Simple CNN, ResNet-18, and VGG-16 Lite. The mean policy weights (0.533x in Simple CNN, 0.784x in ResNet-18, and 0.633x in VGG-16) denote the extent of the adjustment of the RL module in scaling the effective loss, with ResNet-18 experiencing the most significant modification and, therefore, the highest rate of sample reweighting. Combined, these figures demonstrate that RL is somewhat beneficial or non-beneficial to the finest models as they learn discrete weighting patterns for each architecture.

```
•••  RL-Simple Test Accuracy: 0.830 (Policy Weight: 0.533x)
     RL-ResNet Test Accuracy: 0.839 (Policy Weight: 0.784x)
     RL-VGG Test Accuracy: 0.744 (Policy Weight: 0.633x)
```

Figure 12: *The results of RL on the brain-tumor dataset in the form of the accuracy of the tests and the average weights of the policy of Simple CNN, ResNet-18, and VGG-16 Lite. The policy network uses the loss reweighting of the largest loss, which is ResNet-18, as it achieves the maximum RL-adjusted test accuracy among the three models.*

RL vs. Baseline Result Comparison

We see the comparative results as represented by **Table 03**

Model	*Baseline Accuracy (%)*	*RL Accuracy * (%)*	*Absolute Improvement*	*Average Policy Weight*
Simple CNN	76.80	83.00	+0.0620	0.5335
ResNet - 18	72.00	83.90	+0.1190	0.7843
VGG – 16 Lite	72.10	74.40	+0.0230	0.6330

Table 03: *This shows that the three models all gain the enhancement of the dynamic loss weighting through reinforcement learning, and the test accuracy improves to 83%, 83.9%, and 74.4%, respectively, as the models become Simple CNN, ResNet-18, and VGG-16 Lite. The biggest absolute increase comes with the ResNet-18 (+0.119), and the average policy weight (~0.7843) is higher, which means that the policy network is placing more emphasis on its loss reweighting choices. Overall, the table confirms that RL policy is always beneficial, at least in enhancing baseline performance, but not damaging to individual architecture with varying weighting strengths.*

Discussion

The findings of the experiment prove that the suggested CNN-RL framework can deliver competitive results on a somewhat demanding binary brain-tumor MRI dataset that can be run on a CPU-only platform. The three baseline models are complementary to each other: Simple CNN shows the most balanced trade-off between sensitivity and specificity, ResNet-18 can take advantage of transfer learning and converges fastest with high tumor recall, and VGG-16 Lite prefers to be sensitive and have more false positives. The inspection of learning curves suggests that not all architectures overfit critically in the selected epoch budgets, and the hyperparameter choices (64 x 64 grayscale images, Adam optimizer using an initial learning rate of 0.001, small epoch numbers) seem to be suitable to train on augmented MR images in a reasonably short time. The metric-correlation analysis confirms that accuracy, recall, F1-score, and ROC-AUC trend in the same direction, but sensitivity and specificity are at odds with each other, highlighting the fact that when optimizing such systems, miss-tumor reduction should be made a priority.

The proposed solution to the clinical priority of developing a policy-based reinforcement-learning module to encourage dynamic loss weighting can be useful to solve this clinical priority by directly penalizing false negatives more severely than other types of errors and up-weighting hard tumor slices in the training process. In all three backbones, the variants with RL enhance sensitivity and ROC-AUC with only slightly different overall accuracy, which proves that the policy network can learn to use the capacity of the classifier on clinically important edge cases instead of on easy normal images. This design is especially applicable in neuro-oncology, in which a false negative on the presence of an early-stage or subtle lesion is far more detrimental than an apparent scan that will be reviewed [11]. Practically, this type of system might be implemented as a clinical decision-support system that triages MRI diagnosis, indicates high-risk brain slices or examinations to be reviewed by a neuroradiologist, and gives a per-case confidence rating; prior studies have demonstrated that analogous deep-learning assistance systems can increase the accuracy and consistency of radiologist diagnoses and grading of brain tumors. Applied in this supportive mode, where clinicians still maintain ultimate control, the framework can decrease reporting time, normalize sensitivity across readers, and increase access to high-quality tumor detection in environments that lack subspecialty expertise.

Limitations and Future Scope

Meanwhile, there are several limitations that should be taken into consideration. The experiments that are currently being done can only be considered as slice-level, binary presence/absence classification of one augmented public dataset, and 3-D context, tumor subtype, and longitudinal progression are not explicitly modeled, all of which are relevant in the real-world brain tumor treatment planning. The models themselves, such as the RL policy, are also largely black boxes; unless they use integrated explainability (such as saliency maps or attention overlays), clinicians can reasonably be confused to believe or act on automated predictions, particularly on the borderline cases. Lastly, multi-center, multi-scanner cohort external validation must be done before clinical implementation since multi-center performance could deteriorate when the image acquisition protocols or patient groups do not match the training cases [12]. Resolving these problems, such as multi-class or segmentation problems, using explainable AI methods, and evaluating on heterogeneous cohorts, would make the clinical utility and reliability of this policy-based ensemble learning system even more robust when applied to brain-tumor diagnosis.

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Annexure

Annexure Python Code.docx

Open-Source Machine Learning Framework: PyTorch

Python IDE: Google Colab

Runtime Type: CPU (Could not choose GPU or TPU due to daily usage limitation)

Documentation/Resource help for Code Implementation:

- Data Preprocessing_Scikit_Learn.org
- EDA and Pokemon Image Classification_Medium vlog
- Normalization, Data Split and Validation_github.com
- Shallow CNN
- ResNet Model
- VGG 16 Model
- Pixel histogram_geeksforgeeks.org/python/transformation-operations-on-images
- Github.com/Performance_Evaluation_and_Test
- Pytorch.org/ignite/generated/ignite.metrics.confusion_matrix
- Pytorch.org/t/how-to-use-heatmaps-in-predicting-where-the-classifier-is-focusing-on
- Pytorch.org/tutorials/intermediate/reinforcement_learning.html
- Analyticsvidhya.com/blog/2021/02/incremental-and-reinforced-learning-for-image-classification/